Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)

Munshi Nagar, Andheri (W), Mumbai – 400 058.

End Semester Examination

Dec. 2024

Max. Marks: 100 | **Duration: 180 Min.**

Class: Final Year BTech(CE) | **Semester: VII**

Course Code: CS401 | **Program: BTech(CE)**

Name of the Course: Cryptography and System Security

Instruction:

1) **All questions are compulsory.**
2) **Draw neat diagrams and write equations wherever necessary.**
3) **Assume suitable data if necessary but justify the same.**

Course Outcomes: At the end of the course, student will be able to:

(1) Describe the different types of the cryptographic algorithms to secure information.
(2) Apply different cryptographic techniques to solve security-related problems
(3) Create a message digest from data to authenticate authorized user
(4) Use system security practices

Q. No.	Question	Max. Marks	CO-BL
Q. 1A	It can be shown that the Hill cipher with the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ requires that (ad bc) is relatively prime to 26; that is the only common positive factor of (ad bc) and 26 is 1. Thus, if (ad bc) = 13 or is even, the matrix is not allowed. Determine and then justify the number of different (good) keys there are for a 2 x 2 Hill cipher without counting them one by one, using the following steps: a. Find the number of matrices whose determinant is even because one or both rows are even. (A row is "even" if both entries in the row are even.) b. Find the number of matrices whose determinant is even because one or both columns are even. (A column is "even" if both entries in the column are even.) c. Find the number of matrices whose determinant is even because all of the entries are odd. d. Taking into account overlaps, find the total number of matrices whose determinant is even. e. Find the number of matrices whose determinant is a multiple of 13 because the first column is a multiple of 13. f. Find the number of matrices whose determinant is a multiple of 13 where the first column is not a multiple of 13 but the second column is a multiple of the first modulo 13. g. Find the total number of matrices whose determinant is a multiple of 13. h. Find the number of matrices whose determinant is a multiple of 26 because they fit case (a) and (e). (b) and (e). (c) and (e). (a) and (f). And so on ... i. Find the total number of matrices whose determinant is neither a multiple of 2 nor a multiple of 13.	10	CO1-BL5

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Q. 1B	Let π be a permutation of the integers $0, 1, 2, \ldots, (2^n - 1)$ such that $\pi(m)$ gives the permuted value of $m, 0 \neq m \leq 2^n$. Put another way, π maps the set of n-bit integers into itself and no two integers map into the same integer. DES is such a permutation for 64-bit integers. We say that π has a fixed point at m if $\pi(m) = m$. That is, if π is an encryption mapping, then a fixed point corresponds to a message that encrypts to itself. We are interested in the probability that π has no fixed points. Show the somewhat unexpected result that over 60% of mappings will have at least one fixed point.	**10**	CO1-BL4
Q. 2A	A Man-in-the-middle attack can be possible on the Diffie-Hellman key exchange protocol in which the adversary generates two public-private key pairs for the attack. Could the same attack be accomplished with one pair? Show the detailed exchange between legitimate and adversary, if answer is yes. Further show an example of A Man-in-the-middle attack using one pair.	**10**	CO1-BL4
Q. 2B	Perform encryption and decryption using the RSA algorithm for the following: a) p = 3; q = 11, e = 7; M = 5 b) p = 5; q = 11, e = 3; M = 9 c) p = 7; q = 11, e = 17; M = 8 d) p = 11; q = 13, e = 11; M = 7 e) p = 17; q = 31, e = 7; M = 2.	**10**	CO2-BL3
Q. 3A	Discuss Kerberos protocol and show all message exchanges of three servers in Kerberos Protocol. What is the purpose of the Kerberos protocol in network security, and how does it prevent replay attacks?	**10**	CO2,3-BL2
Q. 3B	In what ways can a hash value be secured so as to provide message authentication. Discuss any five ways in detail. **OR** Show the process of creation of 80-word input sequence for SHA-512. Compare SHA-1 and SHA-256 in detail.	**10**	CO3,4-BL3
Q. 4A	What content type in S/MIME provides the following security services: (a) Confidentiality (b) Message Integrity (c) Authentication (d) Nonrepudation (e) Combination of (a) and (b) (f) Combination of (a) and (c) (g) Combination of (a), (b) and (c) (h) Combination of (a), (b), (c) and (d)	**10**	CO2,3, 4-BL3
Q. 4B	a) List four techniques used by firewalls to control access and enforce a security policy. Discuss each technique in detail. b) What are some weaknesses of a packet-filtering router?	**10**	CO4-BL2

Q. 5A	List phases of IKE and the goal of each phase. Define ISAKMP and its relation to IDE. List all ISAKMP payload types and the purpose of each type.	**10**	CO3,4-BL2
Q. 5B	Most protocols have a formula to show the calculations of cryptographic secretes and hashes. For example, in SSL, the calculations of the master secret is as follows (concatenation is designated by a bar i.e. |): Master Secret= MD5(pre-master | SHA-1 ("A" | pre-master | CR | SR)) | MD5(pre-master | SHA-1 ("BB" | pre-master | CR | SR)) | MD5(pre-master | SHA-1 ("CCC" | pre-master | CR | SR)) Show the calculations and formula for the following: a) Key material in SSL b) MAC in SSL c) Hash calculation for CertificateVerify message in SSL d) Hash calculation for Finished message in SSL e) Data expansion in TLS	**10**	CO3,4-BL3